

# Chapter 157 — Diagnosis and Staging of Prostate Cancer

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## Presentation

- **Early prostate cancer is rarely symptomatic**; symptoms imply locally advanced or metastatic disease.
- **Locally advanced**: obstructive urinary symptoms, ureteric obstruction (pain, renal failure), haematuria, reduced ejaculate volume, rarely erectile dysfunction.
- **Metastatic**: bone pain, pathological fracture, lower-limb oedema; less often malignant retroperitoneal fibrosis, paraneoplastic syndromes, disseminated intravascular coagulation (DIC), paralysis from spinal cord compression.
- Such presentations are **now uncommon because of screening** with prostate-specific antigen (PSA) and digital rectal examination (DRE); diagnosis is usually **histological, by transrectal ultrasound (TRUS)-guided needle biopsy**, before symptoms.
- **Staging aims** to define disease extent and risk for management and prognosis, using PSA, DRE, biopsy grade and volume, and imaging (locoregional extent, exclusion of metastases).

## DIAGNOSIS

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### Screening

#### General Concepts of Screening

- **Screening** tests healthy asymptomatic populations; **diagnosis** identifies disease in those with signs or symptoms.
- Goal: better outcomes through earlier treatment — but screening carries **harm from overdiagnosis and overtreatment**; early diagnosis is not invariably beneficial.
- Screening **reduces advanced disease at diagnosis** and, in more recent data, **prostate cancer mortality**; the benefit-harm balance remains contested and depends on the individual's characteristics and preferences.
- **Since widespread PSA testing (early 1990s)**: prostate cancer mortality has fallen by more than 40% and the proportion of advanced-stage disease at diagnosis by about 75% (Surveillance, Epidemiology, and End Results [SEER] data); modelling attributes a substantial share of the mortality fall to screening.
- Randomised trials comparing screened with unscreened populations are the **highest level of evidence**.

#### Randomised Trials

- Both trials began in **1993**; primary endpoint **prostate cancer-specific mortality**.

**SCREENING TRIALS: ERSPEC VS PLCO**

| Feature                     | European Randomised Study of Screening for Prostate Cancer (ERSPC)            | Prostate, Lung, Colorectal and Ovarian (PLCO) trial, National Cancer Institute                                                    |
|-----------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Follow-up (median)          | 16 years                                                                      | 17 years                                                                                                                          |
| Cancer-specific mortality   | <b>20% relative reduction</b>                                                 | <b>No difference</b>                                                                                                              |
| Other effects               | Less high-grade and locally advanced/metastatic disease with serial screening | —                                                                                                                                 |
| Cost of one death prevented | <b>570 invited to screening; 18 extra diagnoses (at 16 years)</b>             | —                                                                                                                                 |
| Why results differ          | —                                                                             | <b>Contamination</b> (high PSA testing in control arm before and during trial); <b>lower power</b> (about threefold fewer events) |

- The table tests **why two trials of one question disagree**: PLCO was effectively screening versus less screening, so a null result is not evidence that screening fails.
- Both trials demonstrate **overdiagnosis** (cancers that would otherwise never be detected) and **overtreatment**.
- **Overtreatment is most concerning above 65 years**: a randomised trial of surgery versus watchful waiting showed minimal treatment benefit in older adults; mean age at diagnosis is about 66, so the risk is high.

**Specialty Group Recommendations****SCREENING RECOMMENDATIONS BY GUIDELINE BODY**

| Body                                            | Recommendation                                                                                                                                                                      |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US Preventive Services Task Force (USPSTF) 2008 | Insufficient evidence $\leq 75$ years; <b>not recommended</b> $\geq 75$                                                                                                             |
| USPSTF 2011 draft / 2012 final                  | <b>Grade D — against screening</b>                                                                                                                                                  |
| USPSTF 2017 draft / 2018 final                  | <b>Grade C, 55–69 years, shared decision-making</b> ; not recommended $\geq 70$ ; rising use of active surveillance has reduced screening harms                                     |
| American College of Preventive Medicine         | Insufficient evidence for routine population screening (PSA and DRE); informed decision                                                                                             |
| American Cancer Society                         | Life expectancy $\geq 10$ years: informed decision <b>from 50 (average risk); earlier if higher risk</b> — Black people, first-degree relative (father or brother) diagnosed $< 65$ |
| American Urological Association (AUA)           | <b>Shared decision-making 55–69; 2-yearly interval</b> (ERSPC-based) preserves benefit, reduces harm                                                                                |
| National Comprehensive Cancer Network (NCCN)    | <b>Baseline PSA and risk–benefit discussion at 45</b> ; retest interval by PSA result (1–4 years)                                                                                   |

- Pattern: **no body recommends unselected population screening**; all converge on **informed or shared decision-making**, differing on start age (45, 50, 55) and stop age (70, 75).
- Trap: USPSTF moved **D (2012) → C (2018)**; the current grade is C, not D.
- Optimal start and stop ages and interval **will stay debated** — further randomised trials are logistically unlikely.
- **Opportunistic screening remains prevalent** in the United States; adults attending periodic health examinations should be told the benefits and risks of PSA testing.

**Concept** — Screening reduces mortality but at the price of overdiagnosis; **every recommendation is an attempt to shift that balance** by selecting age, interval and patient preference, not by denying the benefit.

## Diagnostic Modalities

### Digital Rectal Examination

- Before PSA, **DRE was the sole detection tool**; it has **only fair reproducibility**, even among experienced examiners, and **misses many early cancers**.
- **Limited screening value at PSA <3.0 ng/mL**; some evidence that DRE does not reduce mortality, and some bodies advise against it as a routine screen.
- Drawbacks: blood-only testing lets **reflex tests run on the same sample**; DRE needs a separate attendance (cost, logistics) and **deters some from screening**.
- Nonetheless **most urologists combine PSA and DRE** — an abnormal DRE carries real cancer risk and the examination is simple; many **nomograms incorporate DRE**.

### Prostate-Specific Antigen

- **Human kallikrein gene family** serine protease; secreted into seminal fluid at milligram-per-millilitre concentrations; circulates **bound (complexed) and unbound (free)**, both measurable by US Food and Drug Administration (FDA)-approved assays.

#### Factors influencing PSA.

- **Varies with age, race and prostate volume**; Black people without cancer have higher PSA than White people.
- **PSA rises about 4% per millilitre of prostate volume**; volume explains about 30% and age about 5% of PSA variance.
- **Androgen-dependent**: detectable from puberty as luteinising hormone and testosterone rise; **hypogonadal men may have falsely low PSA** that masks cancer.
- **Assay differences** between serial measurements cause pseudo-acceleration or pseudo-deceleration; genetic factors may justify genetically adjusted PSA.
- **Prostate disease is the single most important determinant** — cancer, benign prostatic hyperplasia (BPH), prostatitis; PSA is **organ-specific, not cancer-specific**, and not all prostate disease raises it.
- Mechanism: **disruption of prostatic architecture** lets PSA reach the circulation — disease or manipulation (massage, biopsy, transurethral resection).
- **DRE causes only a slight rise, within assay error** — rarely a false positive.
- **Ejaculation**: conflicting data; **repeat after 48 hours' abstinence** when PSA is minimally raised.

#### FACTORS THAT RAISE VS LOWER SERUM PSA

| Raise PSA                                         | Lower PSA                                                                                                                                                                        |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prostate cancer, BPH, prostatitis                 | Hypogonadism (low testosterone)                                                                                                                                                  |
| Prostate massage, biopsy, transurethral resection | Obesity (probably haemodilution); statins                                                                                                                                        |
| Increasing age and prostate volume                | <b>5<math>\alpha</math>-reductase inhibitors</b> (finasteride, dutasteride)                                                                                                      |
| Black race                                        | BPH surgery (removes transition zone — "resets" baseline)                                                                                                                        |
| DRE — negligible, within assay error              | Prostate cancer treatment: androgen deprivation (luteinising hormone-releasing hormone agonists, orchidectomy), radiotherapy, radical prostatectomy, ablation (e.g. cryotherapy) |

- **5 $\alpha$ -reductase inhibitors** lower PSA whether type 2-selective (**finasteride**) or dual type 1 and 2 (**dutasteride**); **finasteride 1 mg** (androgenic alopecia) lowers PSA as much as the 5-mg BPH dose.
  - Sequence: **baseline PSA before starting → serial PSA → correct for drug effect**.

- **Double PSA after  $\geq 12$  months**; alternatively  $\times 2.3$  after 2 years,  $\times 2.5$  after 7 years, because response is variable.
- Alternative: treat the **on-treatment nadir as the new baseline** and investigate any subsequent rise.
- Always interpret PSA against **age, urinary tract infection or prostate disease, recent procedures and prostate-directed treatment**.

#### Clinical use for diagnosis.

- FDA approved PSA for **early detection in 1994**; initial assays measured free PSA (fPSA) plus PSA complexed (cPSA) to  $\alpha 1$ -antichymotrypsin — together **total PSA (tPSA)**.
- PSA testing **increases detection and detects cancers more likely to be organ-confined**.
- **Future cancer risk and present biopsy yield both rise incrementally with PSA**, across its full range (Table 157.1).
- Baseline PSA predicts later detection; the **Malmö Preventive Project** showed **PSA in the 40s predicts metastasis and death >25 years later**.

**TABLE 157.1 PROSTATE CANCER DETECTION AS A FUNCTION OF SERUM PROSTATE-SPECIFIC ANTIGEN LEVEL AND DIGITAL RECTAL EXAMINATION FINDINGS IN CONTEMPORARY SERIES**

| PSA (ng/mL) | DRE                         | Detection rate in screened (%) | Yield on biopsy (%) | High-grade (Gleason $\geq 7$ ) on biopsy (%) |
|-------------|-----------------------------|--------------------------------|---------------------|----------------------------------------------|
| 0–1         | Non-suspicious              | —                              | 8.8                 | 0.9                                          |
| 1–2         | Non-suspicious              | —                              | 17.0                | 2.0                                          |
| 0–2         | Non-suspicious / suspicious | — / 0.7                        | 12 / 8              | 1.4                                          |
| 2–4         | Non-suspicious / suspicious | — / 2                          | 15–25 / 21          | 5.2                                          |
| <4          | Non-suspicious / suspicious | — / 1–3                        | 15 / 13–17          | 2.3                                          |
| >4          | Non-suspicious / suspicious | 14 / 14–38                     | 23–38 / 55–63       | 5.8 / 20.6                                   |
| 4–10        | Non-suspicious / suspicious | 11 / 11–27                     | 17–32 / 45–51       | 4.1 / 11.7                                   |
| >10         | Non-suspicious / suspicious | 41 / 31–76                     | 43–65 / 70–90       | 19.4 / 50.5                                  |

Detection rate = cancers  $\div$  all screened. Yield = cancers  $\div$  all biopsied (positive predictive value of PSA with non-suspicious DRE, or of suspicious DRE at that PSA). Data from Andriole et al., 2005; Catalona et al., 1998; Crawford et al., 1996; Schröder et al., 1998; Thompson et al., 2004.

- Purpose: shows **risk is continuous** — **there is no PSA level below which cancer is absent** (about 1 in 11 biopsied at PSA 0–1 ng/mL has cancer).
- Pattern: yield and high-grade rate climb steeply with PSA; **a suspicious DRE multiplies risk at every PSA band**, most strikingly above 10 ng/mL (high-grade rate about half).
- Trap: **detection rate (per screened) and yield (per biopsied) are different denominators** — do not compare them directly.

#### Triggers for prostate biopsy.

- PSA threshold for biopsy is **controversial**; dichotomising PSA into "normal" and "abnormal" discards information below the cut-off — the **Prostate Cancer Prevention Trial (PCPT)** showed risk is **continuous**, and some advise against calling PSA "elevated" or "abnormal".

| Higher threshold                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Lower threshold                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Risks <b>missing significant cancers</b> during the window for cure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>More unnecessary biopsies and overdiagnosis</b> |
| <ul style="list-style-type: none"> <li>• <b>4 ng/mL</b> — original FDA upper limit of normal; many clinicians adopted <b>2.5–3 ng/mL</b>; <b>most ERSPC centres used 3 ng/mL</b>; a <b>higher cut-off (e.g. 10 ng/mL) &gt;70 years</b> has been suggested to reduce harm.</li> <li>• <b>Repeat a suspicious PSA before biopsy</b> — physiological fluctuation causes false positives.</li> <li>• <b>Choosing Wisely: no empirical antibiotics</b> for a raised PSA without urinary symptoms.</li> <li>• Combine PSA with <b>family history, race and DRE</b>.</li> </ul> |                                                    |

### Prostate-Specific Antigen Derivatives and Molecular Forms

- Refinements aim to **improve specificity** by adjusting for volume, rate of change, or molecular form.

#### PSA DERIVATIVES AND MOLECULAR FORMS

| Measure                            | Definition / basis                                                                               | Cut-off                                                      | Use and limits                                                                                                                 |
|------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>PSA density (PSAD)</b>          | PSA ÷ prostate volume (ultrasound or MRI); variants: complexed PSAD, PSA transition zone density | <b>≥0.15</b> → biopsy (PSA 4–10, normal DRE)                 | Offsets BPH; not confirmed in all studies; counselling at 4–10 ng/mL, repeat biopsy, <b>active surveillance eligibility</b>    |
| <b>PSA velocity (PSAV)</b>         | PSA change over elapsed time                                                                     | <b>&gt;0.75 ng/mL/yr</b> (PSA 4–10); <b>0.4</b> at lower PSA | Some studies show no gain over single PSA; varies with calculation method and timing                                           |
| <b>Percentage free PSA (%fPSA)</b> | Cancer → more complexed, less free PSA                                                           | <b>≤25%</b> (PSA 4–10, benign DRE)                           | Detected 95% of cancers, avoided 20% of biopsies; <b>matches PSAD 0.078 without volume measurement</b> ; unclear below 4 ng/mL |
| <b>Complexed PSA (cPSA)</b>        | PSA bound to proteins                                                                            | —                                                            | At high sensitivity: <b>more specific than tPSA, equal to %fPSA</b> (PSA 4–10 and 2.6–4.0)                                     |
| <b>Pro-PSA (proPSA)</b>            | Precursor form from luminal epithelium                                                           | —                                                            | Higher proportion in cancer; may help at PSA 2–10; not always better than %fPSA                                                |
| <b>Benign PSA (bPSA)</b>           | Cleaved inactive fPSA                                                                            | —                                                            | <b>BPH marker</b> — found in nodular BPH tissue                                                                                |
| <b>Intact PSA (iPSA)</b>           | Inactive, uncomplexed fPSA                                                                       | —                                                            | Component of 4Kscore                                                                                                           |
| <b>Human kallikrein 2 (hK2)</b>    | Related kallikrein serine protease; higher in poorly differentiated cancer                       | —                                                            | hK2:fPSA ratio inconsistent for detection; <b>correlates with grade and volume</b> — post-diagnosis risk                       |

- **Transition zone** — the zone of BPH origin — is the main source of serum PSA in men without cancer; this is why volume-adjusted measures and free PSA both work: **BPH raises free PSA and volume, cancer raises complexed PSA** (differential isoform expression, transition vs peripheral zone).
- **%fPSA** rises with age and prostate volume, falls as tPSA rises, varies by assay; **not materially altered by race or 5α-reductase inhibitors**; optimal cut-off depends on prostate size.
- **PSAV and lethal disease: >0.35 ng/mL/yr** 10–15 years before diagnosis → about fivefold risk of life-threatening cancer a decade later; **>2 ng/mL/yr** in the year before diagnosis → cancer-specific death after prostatectomy or radiotherapy.
- Contrast: a systematic review found **pre-treatment PSAV adds nothing to PSA**; other studies show kinetics add predictive value for overall and high-risk disease.

**Multiplex tests.**

- **Prostate Health Index (phi)** — FDA-approved; **tPSA + fPSA + [-2]proPSA**; outperforms tPSA and %fPSA for cancer and high-grade disease in prospective multicentre trials.
- **4Kscore** — **tPSA + fPSA + iPSA + hK2**; predicts clinically significant cancer.

**Other Markers****NON-PSA-DERIVATIVE BIOMARKERS FOR BIOPSY SELECTION**

| Test                                 | Sample                 | Basis                                                                 | Role / bottom line                                                                                      |
|--------------------------------------|------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>IsoPSA</b>                        | Blood                  | Cancer-specific structural PSA isoforms (aqueous two-phase partition) | Validated ≥50 years, PSA ≥4 ng/mL; <b>beats tPSA and %fPSA</b> for any and high-grade cancer            |
| <b>SelectMDx</b>                     | Urine                  | Two-gene mRNA ( <i>HOXC6</i> , <i>DLX1</i> )                          | Biopsy-naïve: avoids a substantial share of biopsies, limits low-grade detection, misses few high-grade |
| <b>ExoDx</b>                         | Urine                  | Exosomal three-gene RNA                                               | Separates Gleason ≥7 from Gleason 6 and benign; avoids biopsies, rarely misses pattern 4                |
| <b>Prostate cancer gene 3 (PCA3)</b> | Urine                  | Non-coding prostate-specific mRNA overexpressed in cancer             | Predicts positive <b>initial or repeat</b> biopsy; <b>conflicting link to aggressiveness</b>            |
| <b>TMPRSS2:ERG</b>                   | Urine                  | Gene fusion (cf. <i>BCR-ABL</i> in leukaemia)                         | Higher cancer specificity; conflicting link to aggressiveness                                           |
| <b>MyProstateScore (MPS)</b>         | Urine + blood          | tPSA + PCA3 + <i>TMPRSS2:ERG</i>                                      | About half avoided immediate biopsy                                                                     |
| <b>ConfirmMDx</b>                    | Tissue (benign biopsy) | Multiplex DNA methylation                                             | <b>After negative biopsy:</b> negative predictive value >90%; avoids unwarranted repeat biopsy          |

- What the table tests: **matching the test to the clinical moment** — blood and urine tests for the first biopsy decision; PCA3 and ConfirmMDx for the **repeat-biopsy** decision.
- **tPSA alone rarely triggers biopsy** — a worrying tPSA is followed by serum and/or urine biomarkers with age, family history and race.

**Concept** — PSA is **sensitive but organ-specific, not cancer-specific**; every derivative, isoform and biomarker in this section exists to recover specificity — separating BPH-driven from cancer-driven PSA — and so reduce unnecessary biopsy.

**STAGING****General Concepts of Staging**

- **Clinical staging** uses pre-treatment parameters to predict extent for prognosis and treatment choice: **DRE** (clinical T stage), PSA and derivatives, needle biopsy features, imaging.

**Clinical Versus Pathological Staging**

| Clinical stage                           | Pathological stage                                                                        |
|------------------------------------------|-------------------------------------------------------------------------------------------|
| Pre-treatment: DRE, PSA, biopsy, imaging | After prostatectomy: histology of prostate, seminal vesicles, pelvic nodes (if dissected) |
| Less accurate                            | <b>More accurate; better outcome predictor</b>                                            |

- **Biochemical recurrence-free and cancer-specific survival are both inversely related to pathological stage.**

- **Pathological predictors after radical prostatectomy:** tumour grade, surgical margin status, extracapsular disease, seminal vesicle invasion, pelvic lymph node involvement.

## Classifications

- **Whitmore–Jewett** staging is now historical; clinical staging uses **tumour, node, metastasis (TNM)**, adopted by the American Joint Committee on Cancer in **1975**.
- Current version combines **TNM + PSA + grade group 1–5** into **prognostic stage groups**.

TABLE 157.2 2017 TNM CLINICAL STAGING SYSTEMS FOR PROSTATE CANCER

| Stage     | Definition                                                                                                                                              |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| TX / T0   | Primary cannot be assessed / no evidence of primary                                                                                                     |
| <b>T1</b> | <b>Not palpable:</b> T1a incidental in $\leq 5\%$ of resected tissue; T1b incidental in $> 5\%$ ; <b>T1c found on needle biopsy</b> (one or both sides) |
| <b>T2</b> | <b>Palpable, confined to prostate:</b> T2a $\leq$ half of one side; T2b $>$ half of one side, not both; T2c both sides                                  |
| <b>T3</b> | Extraprostatic, not fixed: T3a extraprostatic extension (uni- or bilateral); <b>T3b seminal vesicle invasion</b>                                        |
| <b>T4</b> | <b>Fixed or invading adjacent structures other than seminal vesicles</b> — external sphincter, rectum, bladder, levator muscles, pelvic wall            |
| N         | NX not assessed; N0 none; <b>N1 regional node(s)</b>                                                                                                    |
| M         | MX not assessed; M0 none; M1: <b>M1a non-regional nodes; M1b bone; M1c other sites <math>\pm</math> bone</b>                                            |

- Logic: **T1 is defined by how found (not felt), T2 by what is felt;** the 5% cut-off separates T1a from T1b.
- Traps: **seminal vesicle invasion is T3b, not T4; non-regional nodes are M1a, not N1;** a biopsy-detected impalpable cancer is **T1c regardless of bilaterality**.

## Prediction of Tumour Extent

### Prostate-Specific Antigen

- PSA correlates directly with **pathological stage and extent** (its relation to volume is disputed) but **cannot predict extent for an individual:** overlap between stages, variable BPH contribution, and **poorly differentiated tumours make less PSA per gram**.
- **Organ-confined disease by PSA:** about 80% at  $< 4$ ; 66% at 4–10;  $< 50\%$  at  $> 10$  ng/mL.
- **Nodal involvement:** about 20% at PSA  $> 20$ ; 75% at  $> 50$  ng/mL.
- fPSA, hK2, proPSA, PSAD, PSAV and newer tests have been studied as grade and extent predictors; **prostatic acid phosphatase** has been **virtually abandoned** because PSA correlates better.

### Digital Rectal Examination

- Defines **palpability and clinical T stage;** an **abnormal DRE raises risk of grade group 4–5** in screened men.
- **Poor sensitivity and reproducibility  $\rightarrow$  both over- and under-staging;** poor sensitivity with moderate specificity for organ-confined disease; useful only **in combination**.

### Prostate Needle Biopsy

- **Histological grade is the most important biopsy finding;** Gleason is the most used system (1966).
- Gleason sequence: low-power assessment  $\rightarrow$  grade 1–5 for **predominant pattern** + grade for **second most common pattern**  $\rightarrow$  score 2–10.
- **Tertiary pattern** (2005 consensus): on biopsy, **3+4 or 4+3 with tertiary 5  $\rightarrow$  recorded as 3+5 or 4+5**.
- **Scores 2–5 are no longer assigned;** lowest in practice is 6 — confusing for patients and a **barrier to active surveillance**, hence grade groups.

**GLEASON SCORE VS GRADE GROUP**

| Grade group | Gleason score |
|-------------|---------------|
| 1           | ≤6            |
| 2           | 3+4 = 7       |
| 3           | 4+3 = 7       |
| 4           | 4+4 = 8       |
| 5           | 9–10          |

- **Trap: Gleason 7 is two grade groups — 3+4 (GG2) is better than 4+3 (GG3);** the primary pattern matters. Grade groups **simplify, place lowest risk at 1 rather than 6, and predict prognosis accurately.**
- Grade is **not used alone:** number and percentage of positive cores and **intraductal and cribriform patterns** add prognostic information and guide **active surveillance selection.**
- **Seminal vesicle or periprostatic fat on biopsy → worse prognosis;** routine seminal vesicle/capsular biopsy is debated — many reserve it for a **large palpable basal tumour.**

**Combined Use of Pretreatment Parameters**

- **Nomograms and algorithms** combining **T stage (DRE), PSA and Gleason score/grade group** predict extent and long-term outcome **better than any single parameter.**

**D'AMICO RISK GROUPS (FREEDOM FROM DISEASE 10 YEARS AFTER RADICAL PROSTATECTOMY)**

| Risk               | Clinical stage | PSA (ng/mL) | Gleason | 10-year disease-free |
|--------------------|----------------|-------------|---------|----------------------|
| Low (all)          | T1–T2a         | ≤10         | ≤6      | 83%                  |
| Intermediate (any) | T2b            | >10 and ≤20 | 7       | 46%                  |
| High (any)         | T2c            | >20         | 8–10    | 29%                  |

- Logic: **low risk requires every criterion; intermediate and high need only one** — the worst single feature assigns the group.
- Other validated schemes include the **Cancer of the Prostate Risk Assessment (CAPRA)** score; multiparameter risk stratification guides counselling.

**Imaging**

- **Multiparametric MRI (mpMRI):** T2-weighted plus functional **diffusion-weighted and dynamic contrast-enhanced** sequences → better locoregional characterisation; used for **pre-treatment planning and active surveillance candidacy.**
- **MRI-TRUS fusion** enables targeted biopsy; AUA and Society of Abdominal Radiology: **strongly consider high-quality MRI after a prior negative biopsy; suspicious MRI lesion → image-guided targeted biopsy.**
- The chapter's key points go further: **MRI is recommended before prostate biopsy** (improves detection of clinically significant cancer).
- **Risk-based staging:** weigh pre-test probability of metastasis against test sensitivity and specificity.

**STAGING IMAGING BY RISK GROUP**

| Risk group                        | Staging imaging                                                        |
|-----------------------------------|------------------------------------------------------------------------|
| Low or intermediate, asymptomatic | <b>No routine abdominopelvic CT or bone scan</b> — metastasis unlikely |
| High                              | <b>Bone scan + CT or mpMRI; consider PSMA PET</b>                      |

- Nodal disease is rare in screened populations — **staging imaging is overused.**

- **Prostate-specific membrane antigen (PSMA) PET:**  $^{68}\text{Ga}$  PSMA-11 and piflufolastat F-18 ( $^{18}\text{F}$ -DCFPyL) are FDA-approved for **initial staging at high risk of metastasis**; **more sensitive and specific than conventional imaging** for nodal and distant disease; evidence of clinical benefit when conventional imaging is negative remains limited.
- Ultrasound advances — **colour and power Doppler, contrast three-dimensional Doppler, HistoScanning** (computer-aided tissue characterisation) — need larger studies before routine use.

**Concept** — *Imaging yield follows pre-test probability: **the risk group chooses the scan**, which is why low-risk staging scans are wasteful and why the most sensitive tracer is reserved for high-risk disease.*

## Molecular Staging

### TISSUE-BASED GENOMIC TESTS

| Test                 | Measures                                                                                           |
|----------------------|----------------------------------------------------------------------------------------------------|
| Prolaris             | Cell cycle progression (proliferation) gene panel                                                  |
| Oncotype DX Prostate | Quantitative reverse-transcriptase polymerase chain reaction (RT-PCR), <b>17 genes, 5 pathways</b> |
| Decipher             | <b>22 RNA biomarkers</b> , multiple pathways                                                       |

- Predict **adverse pathology, biochemical recurrence, metastasis and cancer death**; may refine standard risk groups but **still need prospective validation** of added value.

### Pelvic Lymphadenectomy

- **No consistent oncological benefit from pelvic lymph node dissection (PLND)**; its value is **staging** — refines risk, informs counselling, guides selective secondary therapy.
- **Risk-stratified:** perform at radical prostatectomy when the **nomogram-predicted probability of nodal disease** exceeds a guideline threshold (thresholds vary); balance against PLND morbidity.

**Concept** — *Staging is prediction under uncertainty: **no single parameter suffices, so parameters are combined** — into risk groups, nomograms and genomic scores — and the predicted risk then decides imaging and nodal dissection.*

### KEY POINTS: DIAGNOSIS AND STAGING OF PROSTATE CANCER

- PSA screening remains controversial; adults at periodic health examinations should be told of the test and **decide on an informed basis**.
- **PSA, DRE, age, race and family history** together predict cancer risk; **prostate disease is the main determinant of PSA**.
- PSA testing **detects more, and more organ-confined, cancers**; future risk and biopsy yield rise incrementally with PSA.
- Most with a raised PSA should have **repeat PSA and/or serum/urine biomarkers before biopsy**.
- **MRI improves detection of clinically significant cancer and is recommended before biopsy**.
- Every diagnosed cancer needs a **risk group**, which directs metastatic imaging and treatment options.
- Survival is inversely related to **pathological stage**; post-prostatectomy predictors: grade, margins, extracapsular disease, seminal vesicle invasion, nodes.